

MISO AFFECTED SYSTEM IMPACT STUDIES

MISO DPP-2018-APR

SOUTHWEST POWER POOL, INC.

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Document Revisions

NO.	Revision	Date	PRD	CHK	APV
1	Draft Report Issued for Review	1/19/2021	WB	TK	TK
2	Revision 1	1/29/2021	WB	WB	WB
3	Revision 2	2/9/2021	WB	WB	WB
4	Revision 3	2/16/2021	WB	WB	WB

Introduction

MISO requested a preliminary Affected System Impact Study from SPP for the DPP-2018-APR cluster for inclusion in Phase II of the DPP process. The purpose of this analysis was to determine the impact of the MISO generation interconnection requests on the SPP transmission system. Additionally, the analysis looked to identify the amount of Interconnection Service available to the projects resulting in no constraints requiring mitigation. This analysis evaluated forty-seven MISO interconnection requests in SPP cluster Groups 9, 12, 13, 15, 16, and 18 with a total generation capacity of 7,072.2 MW. Results from this analysis will be considered preliminary with restudies performed as required at the conclusion of the DISIS-2017-001 and DISIS-2017-002 cluster phases as significant changes to the study assumptions will likely occur¹.

The generation interconnection requests analyzed in this Affected System Impact Study are listed in **Appendix A** by queue number, amount, requested interconnection service type, area, and proposed interconnection point.

The Siemens Power Technologies International PSS/E Version 33.11.0 and PowerGem's TARA 2002 were used for this analysis. SPP provided the following DISIS-2017-001 BASE case models:

- Year 2 (2020) Spring (20G)
- Year 5 (2024) Light (24L)
- Year 5 (2024) Summer Peak (24SP)
- Year 5 (2024) Winter Peak (24WP)

EPE updated power flow cases to reflect the groups under study and developed a total of eighty (80) cases, specifically 40 Base Cases (BC) and 40 Transfer Cases (TC). The power flow analysis was performed to determine if the transmission system can accommodate the injection from the current study cluster generation interconnection requests without violating SPP's transmission planning criteria outlined below in the Study Methodology Criteria Section.

The Affected System Impact Study (ASIS) has been conducted consistent with the Southwest Power Pool (SPP) Open Access Transmission Tariff (OATT), SPP-MISO Joint Operating Agreement (JOA), and SPP Business Practices to determine preliminary impacts to the SPP transmission system.

¹ Significant changes to study assumptions include but are not limited to interconnection request withdrawals and/or changes to higher queued network upgrades included in the base case.

Base Case Model Build and Dispatch

DC Scan

A DC Scan was performed to determine which interconnection requests should be included in the analysis. The Distribution Factor (DF) cut-off criteria used (3%) for DC Scan is based on SPP's Transmission Distribution Factor (TDF) criteria used to identify constraints requiring mitigation reinforcement.

The results of DC Scan analysis revealed that forty-two of the forty-seven projects from DPP-2018-APR met the DC Screening criteria and hence they were included in the analysis. **Table 1** provides a summary of the DC Scan analysis. Distribution factors were calculated using each project as the “sending” system and the MISO Classic² region as the “sink” system. Detailed results for the DC scan can be found in **Table 2**.

Table 1: DC Scan Results by Queue Cluster

Queue Cycle	Region	3% DF Threshold	
		Include	Exclude
Apr-18	Central	12	0
Apr-18	West	30	5

Table 2: Detailed DC Scan Results

Results	
DC Scan Detailed Results	 DC_Screen_Details.xlsx

Base Case Model Review and Grouping

SPP provided the following information to EPE:

1. List of interconnection requests for consideration in the analysis.
2. List of all higher queued interconnection requests.
3. DISIS-2017-001 BASE cases.
4. Latest MISO bench and study cases used for the DPP-2018-APR.
5. Current study network upgrades previously identified by SPP for consideration in the analysis.

The DPP-2018-APR Affected System Impact Study included forty-two (42) generation interconnection requests (“GIRs”) in the MISO footprint. **Appendix A** lists the current study cluster generation interconnection requests included in the study. The BASE cases are ITP cases that have topology added for all queue generators, but all queue generators are not necessarily dispatched. The MISO study generators, including all available collector system data, were added, and kept offline in BASE models listed below:

- Year 2 (2020) Spring (20G)
- Year 5 (2024) Light (24L)
- Year 5 (2024) Summer Peak (24SP)
- Year 5 (2024) Winter Peak (24WP)

² MISO Classic is defined as the PSSE areas of 207, 208, 210, 216, 217, 218, 219, 295, 296, 314, 333, 356, 357, 360, 361, 600, 608, 613, 615, 620, 627, 633, 635, 661, 680, 694, 696, 697, and 698

Higher queued projects were included in the models, including the DISIS-2017-002 cluster or before and DPP Aug-17 and prior. The higher-queued requests included in this study are listed in **Appendix B**. These requests and associated upgrades were added to the BASE models. Note that DISIS-2016-002 Projects and Upgrades were already included in the BASE cases. The DISIS-2017-002 cluster was studied by the Consultant and a group of necessary upgrades was identified. These DISIS -2017-002 and DPP Aug-17 upgrades added to the BASE cases are listed in **Appendix C**.

EPE also identified significant deviations³ between the DISIS-2017-001 BASE cases and the MISO DPP reference cases used for the MISO study. To incorporate updates to the MISO footprint, the MISO shoulder case was used to update MISO representation in the spring and light load seasons. The MISO summer peak case was used to update the MISO representation in the summer and winter peak seasons. The identified topology upgrades in the MISO West and Central areas were added to the BASE models. The summary of the identified transmission topology changes that were incorporated is provided in **Appendix D**.

Grouping

The interconnection requests listed in **Appendix A** and **Appendix B** are grouped into sixteen (16) active regional groups. The SPP groupings are listed in **Table 3** below. The interconnection requests provided in the interconnection request database have been identified as being potentially impactful to the SPP transmission system due to electrical proximity to SPP facilities.

Table 3: All SPP Groupings

Group #	Area	Group #	Area
1	Woodward, OK	10	Southeast OK/Northeast TX
2	Hitchland, OK	12	Northwest AR
3	Spearville, KS	13	Northwest MO
4	Northwest KS	14	South Central OK
6	South TX Panhandle/New Mexico	15	East SD
7	Southwest OK	16	West ND
8	North OK/South Central KS	17	West SD
9	Nebraska	18	East ND

³ Significant deviations would include additions/removals 161 kV+ facilities, changes to R, X, B, and L > 5% for 161 kV+ facilities, and generation differences > 10 MW for interconnection requests under consideration for inclusion in the analysis.

Development of Base Cases (BC Cases)

The number of base and transfer cases required for each impact study depends on the service requested and fuel type of the study units. **Table 4** outlines the number of cases required per seasonal case for each dispatch scenario. **Table 5** describes SPP Dispatch Criteria used for this analysis.

SPP provided the four (4) dispatched seasonal power flow cases. The generation online in the BASE models should reflect the dispatch of the ITP study. The two SPP regional groups (Group 15 and Group 18) then had two types of dispatch for their local generation: High-Variable Energy Resource (HVER) and Network Resource Interconnection Service (NRIS). Additionally, NRIS and LVER dispatches were performed using the entire SPP footprint (Group 00). The groups and the dispatch resulted in 80 cases with unique dispatches, as shown in **Table 4**.

Table 4: SPP Seasons and Cases per Dispatch

Seasonal Case	ERIS HVER	NRIS HVER Cases (G, L)	NRIS HVER Cases (Peak)	ERIS LVER Cases (Peak)
+1 Spring (i.e., 20G)	1 per group	1 per group	--	--
+5 Light Load (i.e., 24L)	1 per group	1 per group	--	--
+5 Summer Peak (i.e., 24SP)	1 per group	--	1 per study	1 per study
+5 Winter Peak (i.e., 24WP)	1 per group	--	1 per study	1 per study
Total Cases	24 Cases	12 Cases	2 Cases	2 Cases
DPP-2018-APR	80 Cases (40 BC/40 TC)			

All in-scope higher queued SPP and MISO generators listed in **Appendix B** were added and dispatched per criteria listed in **Table 5**.

Development of Analysis Cases (TC Cases)

All in-scope higher-queued and current study interconnection requests were dispatched as per criteria listed in **Table 5**. For existing SPP interconnection requests included in the scope, if the existing generation dispatch (P_{GEN}) was greater than the expected GI dispatch criteria, the generation was left as-is. If the existing generation dispatch (P_{GEN}) was less than the expected GI dispatch criteria, it was dispatched up to the defined amount.

Generation adjustments are dispatched against Legacy⁴ conventional generation⁵ in the host TO footprint. For the High Variable Energy Resource (HVER) dispatch scenario, all renewable generation facilities are dispatched to 100% within the studied group and at least 20% outside of the study group. Legacy resources and higher-queued conventional units are used to balance generation changes in the HVER scenarios. For the Light Load condition only, renewables are dispatched at or above 0% capacity outside of the study group. Out-group renewables may also be used to balance generation changes in addition to

⁴ Generators that are found in the SPP footprint in the DISIS BASE cases that do not map to the SPP generator mapping sheet are considered "Legacy" generators.

⁵ Excluding non-adjustable generation such as hydro/run-of-river

legacy and higher-queued conventional units for this scenario. The HVER dispatch scenario was used with all cases including Winter, Summer, Spring, and Light Load DISIS BASE cases.

For the Low Variable Energy Resource (LVER) dispatch scenario, all conventional generation facilities are dispatched to 100%. The code 00 for this scenario represents that the entire SPP footprint is included as being in-group. Legacy resources are used to balance generation changes. The LVER dispatch scenario is utilized in Winter and Summer DISIS BASE cases but only used if there is a conventional resource in the current study.

For the Network Resource (NR) dispatch scenario, the dispatch levels for the renewable and conventional generation facilities are determined based upon the level of system integration being requested (ERIS and NRIS). For Spring and Light Load, dispatches are group based. For Winter and Summer, the entire SPP footprint is considered “in-group” for the study (like the LVER dispatch scenario). Legacy resources are used to balance generation changes.

Each current study interconnection request was included in the power flow analysis models as an equivalent generator dispatched at the applicable percentage of the requested service amount with 0.95 power factor capability. The facility modeling includes explicit representation of equivalent Generator Step-Up (GSU) and main project transformer(s) with impedance data provided in the interconnection request. All equivalent collector system branches and transmission tie-lines shorter than 20 miles in length are modeled as zero-impedance branches.

For the Phase III ASIS for the DPP-2018-APR cluster the NR dispatch will account for MISO’s partial NRIS, where ERIS only capacity will not be dispatched in the NR dispatch scenarios.

Table 5: SPP Dispatch Criteria

Dispatch Type	Season	Service Type	Variable Energy Resource In Group	Variable Energy Resource Out Group	Conventional In Group	Conventional Out Group
ERIS HVER	All	All	100%	20%*	N/A	N/A
ERIS LVER	Peak	All	20%	20%	100%	100%
NRIS	Spring and Light Load	ERIS	80%	20%	N/A	N/A
		NRIS	100%	20%	100%	20%
	Peak	ERIS	20% (solar: 80% in SP)	20% (solar: 80% in SP)	80%	80%
		NRIS	100%	100%	100%	100%
* 0% for DISIS-2017-001 forward						

Study Methodology Criteria

Solve Parameters

The following solution parameters were used:

- Fixed slope decoupled Newton-Raphson
- Tap adjustment – stepping
- Switch shunt adjustments – enable all
- Area interchange disabled
- Adjust phase shift
- Adjust DC taps
- VAR limits – apply immediately
- Must solve within five iterations, three or less is preferred

Thermal Overloads

Network constraints are found by using PowerGEM TARA AC Contingency Calculation (ACCC) analysis on the entire cluster grouping dispatched at the various levels.

For Energy Resource Interconnection Service (ERIS):

For ERIS, thermal overloads are determined for system intact (n-0) (greater than or equal to 100% of Rate A - normal) and for contingency (N-1) (greater than or equal to 100% of Rate B – emergency) conditions.

The overloads are then screened to determine which generator interconnection requests have at least

- 3% Distribution Factor (DF) for system intact conditions (n-0),
- 20% DF upon outage-based conditions (N-1), or
- 3% DF on contingent elements that resulted in a non-converged solution.

Non-converged contingencies shall also be considered for limited operation service.

For Network Resource Interconnection Service (NRIS):

Interconnection Requests that requested NRIS are also studied in a separate NRIS analysis to determine if any constraint measured greater than or equal to a 3% DF. If so, these constraints are also considered for transmission reinforcement under NRIS.

Contingencies

The contingency set includes all SPP control area branches and ties 69kV and above, first tier Non-SPP control area branches and ties 115 kV and above, any defined contingencies for these control areas, and generation unit outages for the SPP control areas with SPP reserve share program redispatch.

- All branches, ties, shunts, and generators within the following areas:
 - SPP Internal Areas for 65kV – 999kV facilities:
 - 515 – 546, 640, 641, 642, 645, 650, 652, 659, 998, 999
 - SPP External Areas for 100kV – 999kV facilities:
 - 327, 330, 351, 356, 502-504, 600, 615, 620, 627, 635, 672, 680
- NERC, SPP, and Tier 1 Permanent Contingent Flowgates
- SPP T.O. Specific P1, P2, P4, and P5 TPL-004-1 Contingencies
- SPP T.O. Specific Op Guide Implementation

Monitored Facilities

The monitored elements include all SPP control area branches, ties, and buses 69 kV and above, and all first tier Non-SPP control area branches and ties 69 kV and above. NERC Power Transfer Distribution Flowgates for SPP and first tier Non-SPP control areas are monitored. Additional NERC Flowgates are monitored in second tier or greater Non-SPP control areas. Voltage monitoring was performed for SPP control area buses 69 kV and above.

- All branches (thermal)/ buses(voltage) and ties within the following areas:
 - SPP Internal Areas for 65kV – 999kV facilities:
 - 515 – 546, 640 – 659, 998, 999
- NERC, SPP, and Tier 1 Permanent Monitor Flowgates (thermal)

Voltage

For non-converged power flow solutions that are determined to be caused by a lack of voltage support, appropriate transmission support will be determined to mitigate the constraint.

After all thermal overload and voltage support mitigations are determined; a full ACCC analysis is then performed to determine voltage constraints. The following voltage performance guidelines are used in accordance with the Transmission Owner local planning criteria.

SPP Areas (69kV+):

Transmission Owner	Voltage Criteria (System Intact)	Voltage Criteria (Contingency)
AEPW	0.95 – 1.05 pu	0.92 – 1.05 pu
GRDA		0.90 – 1.05 pu
KACY		
SWPA		
OKGE		
OMPA		
WFEC		
SWPS		
MIDW		
SUNC		
KCPL		
INDN		
SPRM		

NPPD		
WAPA		
WERE L-V		0.93 – 1.05 pu
WERE H-V		0.95 – 1.05 pu
EMDE L-V		0.90 – 1.05 pu
EMDE H-V		0.92 – 1.05 pu
LES		0.90 – 1.05 pu
OPPD		

SPP Buses with more stringent voltage criteria:

Bus Name/Number	Voltage Criteria (System Intact)	Voltage Criteria (Contingency)
TUCO 230kV 525830	0.925 – 1.05 pu	0.925 – 1.05 pu
Wolf Creek 345kV 532797	0.985 – 1.03 pu	0.985 – 1.03 pu
FCS 646251	1.001 – 1.047 pu	1.001 – 1.047 pu

Affected System Areas (115kV+):

Transmission Owner	Voltage Criteria (System Intact)	Voltage Criteria (Contingency)
AECI	0.95 – 1.05 pu	0.90 – 1.05 pu
EES-EAI		
LAGN		
EES		
AMMO		
CLEC		
LAFA		
LEPA		
XEL		
MP		

SMPA		
GRE		0.90 – 1.10 pu
OTP		0.90 – 1.05 pu
OTP-H (115kV+)	0.97 – 1.05 pu	0.92 – 1.10 pu
ALTW	0.95 – 1.05 pu	0.90 – 1.05 pu
MEC		
MDU		
SPC		0.95 – 1.05 pu
DPC		0.90 – 1.05 pu
ALTE		

The constraints identified through the voltage scan are then screened for the following for each interconnection request.

- 3% DF on the contingent element and
- 2% change in pu voltage

Identification of Network Constraints

ERIS Thermal Non-Converged Constraint Identification and Mitigation

There were no ERIS non-converged constraints observed for single contingency (N-1) conditions.

ERIS Thermal System Intact and Contingency Constraint Identification and Mitigation

Several ERIS thermal constraints were identified for single contingency (N-1) conditions. **Table 6** below summarizes ERIS thermal constraints and associated mitigations.

Table 6: ERIS Thermal Constraints

Monitored Facility	Mitigation
Adams to Creston 161 kV Circuit	Reconductor Adams to Creston 161 kV Circuit
Oahe to Sully Buttes 230 kV Circuit	Oahe to Sully Buttes 230 kV Structure Replacement & Terminal Equipment Upgrade
Whitlock to Glenham 230 kV Circuit	Whitlock to Glenham 230 kV Structure Replacement & Terminal Equipment Upgrade
Bison to Hettinger 230 kV Circuit	Upgrade Terminal Equipment Bison to Hettinger 230 kV Circuit

ERIS Voltage Constraint Identification and Mitigation

There were no voltage constraints identified for Groups 9, 12, 13, 15, 16, or 18 ERIS interconnection requests.

NRIS Thermal Non-Converged Constraint Identification and Mitigation

There were no NRIS non-converged constraints observed for single contingency (N-1) conditions.

NRIS Thermal System Intact and Contingency Constraint Identification and Mitigation

Several NRIS thermal constraints were identified for single contingency (N-1) and multiple contingency conditions. **Table 7** below summarizes NRIS thermal constraints and associated mitigations.

Table 7: NRIS Thermal Constraints

Monitored Facility	Mitigation
Cayler to Wisdom 161 kV Circuit	Cayler to Wisdom 161 kV Structure Replacement
Sheffield to Hampton Tap 161 kV Circuit	Rebuild Sheffield to Hampton Tap 161 kV Circuit
Yankton Jct. to NAPA Jct. 115 kV Circuit	Rebuild Yankton Jct. to NAPA Jct. 115 kV Circuit
Roland to ROLANDTP8 69 kV Circuit	Upgrade Terminal Equipment Roland to ROLDANDTP8 69 kV Circuit
ROLDANDTP8 to RADCLIFF8 69 kV Circuit	Upgrade Terminal Equipment ROLDANDTP8 to RADCLIFF8 69 kV Circuit
S3451 to S3454 345 kV Circuit	Upgrade Terminal Equipment S3451 to S3454 345 kV Circuit
S3451 to S3459 345 kV Circuit	Upgrade Terminal Equipment S3451 to S3459 345 kV Circuit
Emery to Sheffield 161 kV Circuit	Rebuild Emery to Sheffield 161 kV Circuit
Franklin to Hampton Tap 161 kV Circuit	Upgrade Terminal Equipment Franklin to Hampton Tap 161 kV Circuit

NRIS Voltage Constraint Identification and Mitigation

There were no voltage constraints identified for Groups 9, 12, 13,15, 16, or 18 NRIS interconnection requests after accounting for network upgrades associated with non-convergence and thermal issues.

Network Upgrade Cost Estimates, Alternative Solutions, and Limited Operation Availability

Network Upgrade Cost Estimates

Preliminary cost estimates provided in this analysis are subject to change.

SPP utilizes the one-year-out spring seasonal model for Variable Energy Resources (VERs). The five-year-out summer peak seasonal model is used for conventional fuel type generators. If both fuel types are being studied, both sets of models are utilized. Project distribution factors on the identified upgrades, under system intact conditions, are used to determine cost allocation. The impact each generation interconnection request has on each upgrade project is weighted by the size of each request. Finally, the costs due by each request for a particular project are then determined by allocating the portion of each request's impact over the impact of all affecting requests.

For example, assume that there are three Generation Interconnection requests, X, Y, and Z that are responsible for the costs of Upgrade Project '1'. Given that their respective power transfer distribution factors (PTDF) for the project have been determined, the cost allocation for Generation Interconnection request 'X' for Upgrade Project 1 is found by the following set of steps and formulas:

- Request X, Upgrade Project 1 = $PTDF\%(X) * MW(X) = X1$
- Request Y, Upgrade Project 1 = $PTDF\%(Y) * MW(Y) = Y1$
- Request Z, Upgrade Project 1 = $PTDF\%(Z) * MW(Z) = Z1$

Allocation of Cost for a particular project:

- Request X's Project 1 Cost Allocation (\$) = $\frac{\text{Network Upgrade Project 1 Cost (\$)} * X1}{X1 + Y1 + Z1}$

Repeat previous for each responsible GI request for each Project.

If the current study interconnection request requires a network upgrade for full interconnection service, the study resource will determine the Limited Operation amount available to the request prior to all required network upgrades being in-service. **Table 8** lists the allocated costs for Network Upgrades assigned to current study projects.

It should be noted that network upgrades associated with higher-queued projects are also considered as contingent upgrades. These facilities have been included in the models for this study and are assumed to be in service. This list may not be all-inclusive. The interconnection customers, at this time, do not have cost responsibility for these facilities but may later be assigned cost if higher-queued customers terminate their generation. The network upgrades associated with higher-queued projects are listed in **Appendix C**.

Table 8: Network Upgrade Cost Estimates

Interconnection Request	Size	ERIS	NRIS	Total	ERIS Total	NRIS Total	Total
J1001	40	\$0	\$34,839	\$34,839	\$21,700,000	\$35,830,142	\$57,530,142
J1024	200	\$12,000,000	\$0	\$12,000,000			
J1025	300	\$0	\$0	\$0			
J1026	400	\$0	\$0	\$0			
J1033	50	\$0	\$0	\$0			
J1034	225	\$0	\$0	\$0			
J1039	50	\$0	\$0	\$0			
J1040	250	\$465,966	\$176,076	\$642,042			
J1045	20	\$0	\$18,588	\$18,588			
J1050	225	\$0	\$2,000,000	\$2,000,000			
J1072	150	\$0	\$74,983	\$74,983			
J1087	200	\$0	\$0	\$0			
J1092	100	\$0	\$57,759	\$57,759			
J1098	40	\$0	\$252,503	\$252,503			
J1105	200	\$0	\$126,630	\$126,630			
J1106	414	\$0	\$323,611	\$323,611			
J1107	200	\$0	\$0	\$0			
J1110	100	\$0	\$4,419,388	\$4,419,388			
J1122	200	\$0	\$0	\$0			
J1124	100	\$0	\$54,356	\$54,356			
J1128	150	\$0	\$7,882,204	\$7,882,204			
J1132	50	\$0	\$0	\$0			
J1140	80	\$0	\$52,363	\$52,363			
J1145	250	\$0	\$0	\$0			
J1164	80	\$0	\$759,941	\$759,941			
J1169	50	\$0	\$1,968,089	\$1,968,089			

Interconnection Request	Size	ERIS	NRIS	Total	ERIS Total	NRIS Total	Total
J1174	165	\$0	\$8,526,060	\$8,526,060			
J1175	165	\$0	\$8,526,060	\$8,526,060			
J1181	200	\$0	\$80,191	\$80,191			
J1182	250	\$0	\$0	\$0			
J1187	151.8	\$318,142	\$102,960	\$421,102			
J952	54	\$8,700,000	\$0	\$8,700,000			
J953	2	\$0	\$0	\$0			
J954	1.4	\$0	\$0	\$0			
J956	200	\$0	\$0	\$0			
J959	150	\$0	\$50,359	\$50,359			
J967	150	\$0	\$74,983	\$74,983			
J975	150	\$215,892	\$0	\$215,892			
J976	300	\$0	\$0	\$0			
J982	300	\$0	\$268,197	\$268,197			
J987	100	\$0	\$0	\$0			
J994	100	\$0	\$0	\$0			

Alternative Solutions

To identify the final network upgrades, numerous combinations and options were tested for effectiveness. The alternative solutions were not chosen due to higher costs.

Table 9: Alternative Solutions

Type	Constraint(s)	Mitigation	Alternative
ERIS	Adams to Creston 161 kV Circuit	Reconductor Adams to Creston 161 kV Circuit	Rebuild Adams to Creston 161 kV Circuit
ERIS	Oahe to Sully Buttes 230 kV Circuit	Oahe to Sully Buttes 230 kV Structure Replacement & Terminal Equipment Upgrade	Rebuild Oahe to Sully Buttes 230 kV Circuit
ERIS	Whitlock to Glenham 230 kV Circuit	Whitlock to Glenham 230 kV Structure Replacement & Terminal Equipment Upgrade	Rebuild Whitlock to Glenham 230 kV Circuit
NRIS	Cayler to Wisdom 161 kV Circuit	Cayler to Wisdom 161 kV Structure Replacement	Rebuild Cayler to Wisdom 161 kV Circuit
NRIS	S3451 to S3454 345 kV Circuit	Upgrade Terminal Equipment S3451 to S3454 345 kV Circuit	Rebuild S3451 to S3454 345 kV Circuit
NRIS	S3451 to S3459 345 kV Circuit	Upgrade Terminal Equipment S3451 to S3459 345 kV Circuit	Rebuild S3451 to S3459 345 kV Circuit

Limited Operation Availability

The results of the Power Flow identified the system constraints that require mitigation. The Limited Operation Analysis evaluated the most limiting of these constraints for each current study request and identifies an amount of available interconnection service.

Power Flow Analysis results included the thermal overload amount, circuit rating, size, and TDF of each current study request. An initial Limited Operation amount is calculated by identifying the impact of each request on each constraint and identifying a reduced size of each request proportional to the thermal constraint that would result in a circuit loading within the applicable rating. The Limited Operation amount is calculated according to the following equation:

$$\text{Limited Operation amount} = \text{Request MW} - \frac{\text{MVA Rating} * (\text{Overload PU} - 1)}{\text{Request TDF}}$$

With the initial Limited Operation amount request sizes applied to the study cases, ACCC is repeated to verify that the thermal constraints are not observed, or the calculation and verification are repeated until all thermal constraints are mitigated.

Power Flow Analysis results for voltage violations are then further mitigated by identifying the contribution of each request and determination of the required impact reduction is conducted and verified through ACCC to determine the Power Flow Analysis Limited Operation amount for each request.

Limited Operation Results are listed below in **Table 10**. While these results are based on the criteria listed in GIP 8.4.3, the Interconnection Customers may request additional scenarios for Limited Operation based on higher-queued Interconnection Requests not being placed in service.

Table 10: Limited Operation Results

Interconnection Request	Group	Service Type	Available MW Before Mitigation	Most Limiting Constraint
J1001	15 E-SD	ERIS	40	None
		NRIS	0	S3451 to S3459 345 kV
J1024	13 NE-KS & NW-MO	ERIS	0	Adams to Creston 161 kV
		NRIS	200	None
J1025	13 NE-KS & NW-MO	ERIS	300	None
		NRIS	300	None
J1026	13 NE-KS & NW-MO	ERIS	400	None
		NRIS	400	None
J1033	12 W-ARK	ERIS	50	None
		NRIS	50	None
J1034	12 W-ARK	ERIS	225	None
		NRIS	225	None
J1039	13 NE-KS & NW-MO	ERIS	50	None
		NRIS	50	None
J1040	18 E-ND	ERIS	0	Bison to Hettinger 230 kV
		NRIS	0	S3451 to S3454 345 kV
J1045	15 E-SD	ERIS	20	None
		NRIS	0	S3451 to S3459 345 kV
J1050	09 NEB	ERIS	225	None
		NRIS	165	Roland to Roland Tap 69 kV

Interconnection Request	Group	Service Type	Available MW Before Mitigation	Most Limiting Constraint
J1072	15 E-SD	ERIS	150	None
		NRIS	0	S3451 to S3459 345 kV
J1087	12 W-ARK	ERIS	200	None
		NRIS	200	None
J1092	15 E-SD	ERIS	100	None
		NRIS	0	S3451 to S3459 345 kV
J1098	15 E-SD	ERIS	40	None
		NRIS	0	Cayler to Wisdom 161 kV
J1105	15 E-SD	ERIS	200	None
		NRIS	0	S3451 to S3459 345 kV
J1106	15 E-SD	ERIS	414	None
		NRIS	0	S3451 to S3459 345 kV
J1107	12 W-ARK	ERIS	200	None
		NRIS	200	None
J1110	15 E-SD	ERIS	100	None
		NRIS	0	Sheffield to Hampton Tap 161 kV
J1122	09 NEB	ERIS	200	None
		NRIS	200	None
J1124	15 E-SD	ERIS	100	None
		NRIS	0	S3451 to S3459 345 kV
J1128	15 E-SD	ERIS	150	None
		NRIS	0	S3451 to S3459 345 kV
J1132	09 NEB	ERIS	50	None
		NRIS	50	None
J1140	18 E-ND	ERIS	80	None
		NRIS	0	S3451 to S3459 345 kV
J1145	13 NE-KS & NW-MO	ERIS	250	None
		NRIS	250	None
J1164	15 E-SD	ERIS	80	None
		NRIS	0	Cayler to Wisdom 161 kV
J1169	18 E-ND	ERIS	50	None
		NRIS	0	S3451 to S3459 345 kV
J1174	15 E-SD	ERIS	165	None
		NRIS	0	S3451 to S3459 345 kV
J1175	15 E-SD	ERIS	165	None
		NRIS	0	S3451 to S3459 345 kV
J1181	15 E-SD	ERIS	200	None
		NRIS	0	S3451 to S3459 345 kV
J1182	13 NE-KS & NW-MO	ERIS	250	None
		NRIS	250	None
J1187	16 W-ND	ERIS	0	Bison to Hettinger 230 kV
		NRIS	0	S3451 to S3454 345 kV
J952	16 W-ND	ERIS	0	Oahe to Sully Buttes 230 kV
J953	09 NEB	ERIS	2	None
		NRIS	2	None

Interconnection Request	Group	Service Type	Available MW Before Mitigation	Most Limiting Constraint
J954	09 NEB	ERIS	1.4	None
		NRIS	1.4	None
J956	13 NE-KS & NW-MO	ERIS	200	None
		NRIS	200	None
J959	15 E-SD	ERIS	150	None
		NRIS	0	S3451 to S3459 345 kV
J967	15 E-SD	ERIS	150	None
		NRIS	0	S3451 to S3459 345 kV
J975	18 E-ND	ERIS	0	Bison to Hettinger 230 kV
J976	13 NE-KS & NW-MO	ERIS	300	None
		NRIS	300	None
J982	15 E-SD	ERIS	300	None
		NRIS	0	S3451 to S3459 345 kV
J987	13 NE-KS & NW-MO	ERIS	100	None
		NRIS	100	None
J994	13 NE-KS & NW-MO	ERIS	100	None
		NRIS	100	None

Power Flow Analysis

The results of the power flow analysis for interconnection requests under study are embedded in **Table 11**.

Table 11: Power Flow Analysis Results

Results	
Thermal Constraints	 APR18_Thermal_Constraints.xlsx
Network Upgrades and Cost Allocation Calculations	 APR18_Network_Upgrades_Cost_Allocation

Conclusion

A power flow analysis was performed to determine the impact of forty-two (42) MISO GIRs on the SPP transmission system. The results of the Power Flow analysis identified the system constraints that require mitigation. The Limited Operation Analysis evaluated the most limiting of these constraints for each current study request and identified an amount of available interconnection service. The minimum cost of interconnecting all new generation interconnection requests included in this analysis is estimated at \$57.530 million. This minimum cost is subject to the higher queued upgrades required for the projects included in the DISIS-2017-002 cluster listed in **Appendix C**. Allocated costs for Network Upgrades are listed in **Table 8**.

The study results identified several ERIS constraints. Subject to an interim Limited Operation Affected System Impact Study full ERIS interconnection service capacity is available for projects J1001, J1025, J1026, J1033, J1034, J1039, J1045, J1050, J1072, J1084, J1087, J1092, J1098, J1105, J1106, J1107, J1110, J1122, J1124, J1128, J1132, J1140, J1145, J1164, J1169, J1174, J1175, J1181, J1182, J953, J956, J956, J959, J967, J976, J982, J987, and J994 as they do not have identified ERIS constraints/upgrades. Projects J1024, J1040, J1187, J952, and J975 require ERIS upgrades to be in service before interconnection service is available.

The study results also identified several NRIS constraints. Subject to an interim Limited Operation Affected System Impact Study full NRIS interconnection service capacity is available for projects J1024, J1025, J1026, J1033, J1034, J1039, J1087, J1107, J1122, J1132, J1145, J1182, J953, J954, J956, J976, J987, and J994 as they do not have identified NRIS constraints/upgrades. Projects J1001, J1040, J1045, J1050, J1072, J1092, J1098, J1105, J1106, J1110, J1124, J1128, J1140, J1164, J1169, J1174, J1175, J1181, J1187, J959, J967, and J982 require NRIS upgrades to be in service before any NRIS Service capacity is available.

Appendix A: Current Study Interconnection Requests

Generation Interconnection Number	Study	Group	Type	G PMAX	SP PMAX	WP PMAX	Service	GEN Area	Point of Interconnection
J1001	DIS-18-1-PQ	15 E-SD	Solar	40	40	40	ER/NR	XEL	BUFFRID7
J1024	DIS-18-1-PQ	13 NE-KS & NW-MO	Wind	200	200	200	ER/NR	MEC	Maryville - Braddyville 161kV Line Tap
J1025	DIS-18-1-PQ	13 NE-KS & NW-MO	Wind	300	300	300	ER/NR	AMMO	7ZACKARY
J1026	DIS-18-1-PQ	13 NE-KS & NW-MO	Wind	400	400	400	ER/NR	AMMO	7SPENCER
J1033	DIS-18-1-PQ	12 W-ARK	Battery	50	50	50	ER/NR	AMMO	Morley - Stoddard 161kV Line Tap
J1034	DIS-18-1-PQ	12 W-ARK	Solar	225	225	225	ER/NR	AMMO	Morley - Stoddard 161kV Line Tap
J1039	DIS-18-1-PQ	13 NE-KS & NW-MO	Battery	50	50	50	ER/NR	AMMO	7MONTGMRY
J1040	DIS-18-1-PQ	18 E-ND	Wind	250	250	250	ER/NR	MDU	WISHEK 4
J1045	DIS-18-1-PQ	15 E-SD	Battery	20	20	20	ER/NR	XEL	FENTON 7
J1050	DIS-18-1-PQ	09 NEB	Wind	225	225	225	ER/NR	ALTW	DOUD 5
J1072	DIS-18-1-PQ	15 E-SD	Solar	150	150	150	ER/NR	XEL	ADAMS 3
J1084	DIS-18-1-PQ	09 NEB	Solar	150	150	150	ER/NR	XEL	ROCK CK3
J1087	DIS-18-1-PQ	12 W-ARK	Solar	200	200	200	ER/NR	AMMO	5KELSO
J1092	DIS-18-1-PQ	15 E-SD	Solar	100	100	100	ER/NR	XEL	THREE LAKES7
J1098	DIS-18-1-PQ	15 E-SD	Solar	40	40	40	ER/NR	ALTW	LKFLDXL3
J1105	DIS-18-1-PQ	15 E-SD	Solar	200	200	200	ER/NR	XEL	HMPT CNR3
J1106	DIS-18-1-PQ	15 E-SD	Wind	414	414	414	ER/NR	XEL	LYON CO 3
J1107	DIS-18-1-PQ	12 W-ARK	Solar	200	200	200	ER/NR	AMMO	7KELSO
J1110	DIS-18-1-PQ	15 E-SD	Solar	100	100	100	ER/NR	SMPMPA	AUSTIN 5
J1122	DIS-18-1-PQ	09 NEB	Wind	200	200	200	ER/NR	MEC	OVRLND 3
J1124	DIS-18-1-PQ	15 E-SD	Solar	100	100	100	ER/NR	XEL	BYRON 3
J1128	DIS-18-1-PQ	15 E-SD	Solar	150	150	150	ER/NR	SMPMPA	HAYWARD L2 5
J1131	DIS-18-1-PQ	09 NEB	Solar	100	100	100	ER/NR	MEC	SB 56 5
J1132	DIS-18-1-PQ	09 NEB	Solar	50	50	50	ER/NR	ALTW	ECRESTN8
J1135	DIS-18-1-PQ	09 NEB	Solar	50	50	50	ER/NR	ALTW	AGENCY J1 8
J1140	DIS-18-1-PQ	18 E-ND	Solar	80	80	80	ER/NR	XEL	PLATTRV7
J1145	DIS-18-1-PQ	13 NE-KS & NW-MO	Solar	250	250	250	ER/NR	AMMO	7MCCREDIE
J1164	DIS-18-1-PQ	15 E-SD	Solar	80	80	80	ER/NR	ALTW	MAGNLIA5
J1169	DIS-18-1-PQ	18 E-ND	Solar	50	50	50	ER/NR	XEL	GRANT 7
J1174	DIS-18-1-PQ	15 E-SD	Solar	165	165	165	ER/NR	ALTW	COLBY3
J1175	DIS-18-1-PQ	15 E-SD	Wind	165	165	165	ER/NR	ALTW	COLBY3
J1181	DIS-18-1-PQ	15 E-SD	Wind	200	200	200	ER/NR	ALTW	HAZLTON3
J1182	DIS-18-1-PQ	13 NE-KS & NW-MO	Solar	250	250	250	ER/NR	AMMO	7ZACKARY
J1187	DIS-18-1-PQ	16 W-ND	Wind	151.8	151.8	151.8	ER/NR	GRE	GRE-STANTON4

MISO Affected System Impact Studies

Generation Interconnection Number	Study	Group	Type	G PMAX	SP PMAX	WP PMAX	Service	GEN Area	Point of Interconnection
J952	DIS-18-1-PQ	16 W-ND	Wind	54	54	54	ER	MDU	MCINTSH7
J953	DIS-18-1-PQ	09 NEB	Diesel	1.83	1.83	2	ER/NR	AMIL	1DUPOFRY2
J954	DIS-18-1-PQ	09 NEB	Solar	1.4	1.4	1.4	ER/NR	AMIL	1DUPOFRY2
J956	DIS-18-1-PQ	13 NE-KS & NW-MO	Solar	200	200	200	ER/NR	AMMO	7SPENCER
J959	DIS-18-1-PQ	15 E-SD	Wind	150	150	150	ER/NR	ALTW	WINDSOR5
J963	DIS-18-1-PQ	09 NEB	Diesel	9	9	9	ER/NR	ALTW	TIPTONM8
J967	DIS-18-1-PQ	15 E-SD	Wind	150	150	150	ER/NR	XEL	ADAMS 3
J975	DIS-18-1-PQ	18 E-ND	Wind	150	150	150	ER	OTP	BUFFALO7
J976	DIS-18-1-PQ	13 NE-KS & NW-MO	Solar	300	300	300	ER/NR	AMMO	7MONTGMRY
J981	DIS-18-1-PQ	09 NEB	Wind	200	200	200	ER/NR	MEC	SUB T HSK 3
J982	DIS-18-1-PQ	15 E-SD	Wind	300	300	300	ER/NR	MEC	PALO ALTO 3
J987	DIS-18-1-PQ	13 NE-KS & NW-MO	Solar	100	100	100	ER/NR	AMMO	5MONTGMRY
J994	DIS-18-1-PQ	13 NE-KS & NW-MO	Solar	100	100	100	ER/NR	AMMO	5GUTHRIE

Appendix B: Higher Queued Interconnection Requests



Higher Queued
Projects.xlsx

Appendix C: Higher Queued Network Upgrades Added to Models



Higher Queued
Network Upgrades.x

Appendix D: MISO BASE Topology Updates included in this Analysis

Description	Area Number	kV
Add 2nd 345-161 kV transformer at Sub 39	635	345/161
New Beehive 345/138 kV Substation	356	345/138
Rebuild Messenger - Carbide 69 kV to double circuit	356	69
New Cardinal - Hickory Creek 345 kV	600	138
Rebuild Triboji - Orleans Tap 69kV line	635	69
New Elisha 115/34.5 kV substation using Hubbard transformer	608	115/34.5
Build 115 kV line from Potato Lake tap to Arago distribution substation.	608	115
Construct a new substation in the Hills area.	627	345/69
Construct a new 3 terminal switch station near Lyon	652	69
New 69 kV line to add the new Marion South sub.	627	69
Install 345/138 kV transformer at Gateway	357	345/138
Replace 161 kV circuit breaker at Bondurant Substation	635	161
New 115 kV substation yard adjacent to the Boswell 230 kV yard. A new 230/115 kV transformer will connect both yards.	608	230/115
New +/-75 MVAR STATCOM	608	115
Build a new 115 kV line from the MP Big Rock substation to the GRE Waldo substation	608	115
Extend existing Forbes - Laskin 115 kV Line (38 Line) up to Hoyt Lakes, removing connection to Laskin	608	115
New 115 Tioga Substation	608	115/23
Construct a second 161 kV line from Sub A to Sub 39.	635	161
Reconfigure the 115 kV near Indiana	600	115
Expansion of the existing Hollydale substation to accommodate serving load at Hollydale at 69 kV on a permanent basis.	600, 615	69/13.8
New 115 kV from WAPA's Miles City #2 substation to MDU's new Miles City SW 115 kV substation.	652, 661	115
New 161 kV Bondurant substation	635	161/13
Upgrade Red Rock TR 9 and associated bus work and equipment to higher rating	600	345/115
Rebuild/replace the existing Glenwood 69-13 kV Substation to improve reliability.	680	69/13
Adds 10MVAR cap bank at Turtle Lake	680	69
Rebuild 2 69kV line feeders into new Clear Lake Substation across the street from existing Clear Lake Substation. Install new 69kV breakers and buswork.	600	69
Rebuild and upgrade 14 miles of 69 kV line to 138 kV design (Altoona - Pierson)	635	138
Replace bus tie 3-4 OCB, and pos S OCB at Maline substation	356	138
Upgrade Hazel Creek TR9 to accommodate the interconnection of J460 - Blazing Star 1.	600	345/230

Description	Area Number	kV
New 345 kV and 230 kV facilities at Big Stone South, Brookings County, and Whapeton to accommodate multiple Generation Interconnection Requests	620	345
Add new 69 kV Path (Maquoketa - Delmar)	627	69
Add new 69 kV path (Taft & Brinsdale - Prairie)	620	69
Add parallel branch and transformer at Zachary	356	345/161
Add parallel transformer at Sub 18	635	161/69
Add series capacitor at Briggs Road	600	345
Add series capacitor at Pillsbury	620	230
Reconfigure 69 kV at Court Tap	627	69
Add new 69 kV path at Blomkest	600	69
Reconfigure Adams 161 kV substation	627	161
Add new 69 kV path at Hector Tap	600	69
Add new transformer at Frontier	620	230/115
Add new Raccoon 345 kV station	635	345
Reconfigure 69 kV at Bryantsburg	627	69
Reconfigure 69 kV at Slakes	627	69
Reconfigure 69 kV at Morrell and Cliff Ave	600	69
Reconfigure 69 kV at IPL Tiffin	627	69
Remove transformer at Poplar Lake	680	69
Add new 69 kV path from Ryan to Manchester	627	69
Reconfigure 69 kV at Huntley	627	69
Reconfigure 118.1 kV at Fort Frances	103	118.1
Remove duplicate modeling of transformers at Hancock	627	69
Add Walters 161 tap and 161/69 transformer	627	161
Remove parallel transformer at Henry County	330	161/69
Reconfigure Hall transformer connections	356	138/34.5
Transformer parameter modifications, including all impedance changes >5%	Multiple	Multiple
Branch parameter modifications, including all impedance changes >5%	Multiple	Multiple
Add, remove, and modify various switched shunts	Multiple	Multiple